

4Corrigendum

Cercis-Fisher

Cercis-Fisher Equivalence: Conditional Equivalence, Not Natural Emergence

SCX

July 2, 2026

1 / Review Verdict

/ Original Claim	/ Verdict
Cercis ² $\propto$ KL = Fisher	“” / Mathematically correct but “natural emergence” is overstated

1.1 / Specific Issues

- (1) **“KL” “KL”** CercisKL $\mathcal{C} \approx 2 \cdot \min_h \text{KL}(P_A \| P_{doh})$ “KL”

Not “average KL” but “minimum KL”. Cercis is the *minimum* KL divergence to the pure-gauge submanifold, not an average.

- (2) “” GeoDist² $\leftrightarrow$ 2KL 8?? SCX

Not “natural emergence” but conditional equivalence. The GeoDist² $\leftrightarrow$ 2KL equivalence depends on 8 strong assumptions (see §??). If any single assumption fails, the equivalence breaks. SCX data is extremely unlikely to form an exponential family.

- (3) GeoDist² 2KL Amari-Chentsov SCX

Complete breakdown outside exponential families. For general statistical manifolds, the difference between GeoDist² and 2KL is controlled by the Amari-Chentsov third-order tensor. This tensor is generically non-zero for SCX data, so the equivalence degrades.

2 / The Eight Assumptions

$$\text{GeoDist}(p, q)^2 = 2\text{KL}(p \| q) + O(\|p - q\|^3)$$

For GeoDist² $= 2\text{KL}(p \| q) + O(\|p - q\|^3)$ to hold rigorously, the following eight assumptions must be simultaneously satisfied. Failure of any single one degrades the equality.

H1: (Exponential Family Assumption).

$\mathcal{P} = \{p(x | \theta)\}$ $T(x)$ $\psi(\theta)$ $p(x | \theta) = h(x) \exp(\langle \theta, T(x) \rangle - \psi(\theta))$ Fisher ψ Hessian KLBregman4.3 / Lemma ??

English: The output distribution family $\mathcal{P}$ must be an exponential family. Only then is the Fisher matrix the Hessian of ψ , and KL divergence equals Bregman divergence.

KL($\theta_1 \| \theta_2$) ψ Bregman Taylor $g_{ij} \Delta \theta^i \Delta \theta^j$ GeoDist² 2KL Amari-Chentsov 6.2 / Open Problem ??

Consequence of failure: Outside exponential families, KL is no longer a Bregman divergence, and the second-order Taylor term is no longer $g_{ij} \Delta \theta^i \Delta \theta^j$. The difference is controlled by the Amari-Chentsov third-order tensor (Open Problem ??).

H2: (Small-Distance / Local Approximation Assumption).

$\theta_p, \theta_q \ O(\|\theta_q - \theta_p\|^3)$ Taylor

English: θ_p and θ_q must be sufficiently close so that third-order and higher remainders $O(\|\theta_q - \theta_p\|^3)$ are negligible. All derivations truncate the Taylor expansion at second order.

large discrepancy $\text{GeoDist}^2 \approx 2\text{KL SCX}$

Consequence of failure: For large discrepancies (common in cross-domain SCX experts), third-order error terms dominate and $\text{GeoDist}^2 \approx 2\text{KL}$ no longer holds.

H3: C^2 (C^2 Differentiability Assumption).

$\theta \mapsto p(x | \theta)$ $x \ C^2$ TaylorFisher

English: The map $\theta \mapsto p(x | \theta)$ must be C^2 for almost all x . This is required for second-order Taylor expansion and for the Fisher matrix as a second derivative.

ReLU Fisher

Consequence of failure: If the density has non-differentiable points (e.g., piecewise distributions from ReLU activations), the Fisher metric is undefined at those points and geodesics are not uniquely determined.

H4: - (Integration-Differentiation Interchangeability Assumption).

$\partial_\theta \int p(x|\theta) d\mu(x) = \int \partial_\theta p(x|\theta) d\mu(x)$ Fisher $\mathbb{E}[\partial_\theta \log p] = 0$

English: The interchange $\partial_\theta \int p = \int \partial_\theta p$ must hold. This is the standard regularity condition ensuring $\mathbb{E}[\partial_\theta \log p] = 0$, which underpins the Fisher information definition.

Fisher

Consequence of failure: If the support depends on θ (e.g., uniform distributions), the standard Fisher information form breaks down.

H5: Fisher (Fisher Full-Rank Assumption).

Fisher $\mathcal{I}(\theta) \ \Theta$ Fisher —

English: The Fisher information matrix $\mathcal{I}(\theta)$ must be positive-definite (full rank) at every point. This is required for (Θ, g) to be a genuine Riemannian manifold — a degenerate metric does not define unique geodesics.

Fisher GeoDist

Consequence of failure: If parameters are functionally dependent (e.g., normalization constraints causing effective dimensionality reduction), the Fisher matrix is singular, geodesics may not exist or be unique, and GeoDist is undefined.

H6: (Small-Connection Approximation Assumption).

$a_e, h_v \ \max_e \text{dist}_{O(d)}(A_e, I_d) < r_0 \ r_0 \ \mathfrak{so}(d)$

English: Edge assignments a_e and gauge transformations h_v must be within the small-connection limit ($\max_e \text{dist}_{O(d)}(A_e, I_d) < r_0$, the matrix logarithm convergence radius). Only then does non-abelian gauge-fixing reduce to linear Hodge theory on $\mathfrak{so}(d)$.

90° Cartan GeoDist

Consequence of failure: Under large rotations (e.g., 90° frame misalignment between cross-modal experts), nonlinear Cartan terms cannot be neglected, and the relationship between GeoDist and Euclidean distance breaks.

H7: (Euclidean Approximation / Degenerate Fisher Assumption).

Cercis $\|r\|^2 = \|r_{\text{harm}}\|^2 + \|r_{\text{coexact}}\|^2$ FisherFisher $\mathcal{I}_{ij}(\theta) = \delta_{ij}$

English: For the Euclidean Cercis definition ($\|r\|^2 = \|r_{\text{harm}}\|^2 + \|r_{\text{coexact}}\|^2$) to coincide with the Fisher definition, the Fisher metric must degenerate to Euclidean: $\mathcal{I}_{ij}(\theta) = \delta_{ij}$. This means all directions on the information manifold have equal weight and zero curvature.

SCXFisher CercisFisher

Consequence of failure: Real SCX parameter spaces almost certainly have non-trivial Fisher curvature. A non-negligible curvature correction exists between Euclidean Cercis and Fisher geodesic distance.

H8: (Distribution Compactness / Bounded Support Assumption).

Θ Taylor

English: All expert output distributions must be sufficiently concentrated on a compact subset of Θ , so that the local Taylor expansion is uniformly valid across all boundary observations and gauge bulk states.

-t

Consequence of failure: Heavy-tailed distributions (e.g., Student’s-t) may place some observations outside the local validity domain, causing uncontrolled growth of third-order error terms on individual edges.

3 / Corrected Claim

3.1 (Chinese Correction)

“CercisFisherCercis² $\propto$ KL = Fisher——”

CercisKL Fisher ——“” Situs **Amari-Chentsov6.2**

Cercis

8(H1–H8)

$$\mathcal{C} = \min_{d_0 h \in \text{im}(d_0)} \text{GeoDist}(P_A, P_{d_0 h})^2 = 2 \min_{d_0 h \in \text{im}(d_0)} \text{KL}(P_A \| P_{d_0 h}) + O(\|\mathbf{a}\|^3).$$

Cercis“” Fisher

(i) (H1)GeoDist²2KL Amari-Chentsov $T_{ijk} = \mathbb{E}[\partial_i \log p \cdot \partial_j \log p \cdot \partial_k \log p]$ T_{ijk} SCX??

(ii) (H2)

(iii) “”——

3.2 English Correction

Original claim (overstated, to be removed):

“The Cercis Score is the geodesic distance under the Fisher information metric. Under the exponential family assumption, Cercis² $\propto$ mean KL divergence = squared Fisher geodesic length. The audit score is not artificially defined — it naturally emerges from the information geometry of the data manifold.”

Corrected claim (conditional equivalence):

Under the exponential family assumption and local approximation, the Cercis Score equals twice the **minimum** KL divergence from the observed edge distribution to the pure-gauge submanifold, which in turn equals (to second order) the squared Fisher geodesic distance. This provides an information-geometric **interpretation** — not a “natural emergence” but a **conditional equivalence** that holds when and only when the Situs manifold admits an exponential

family structure. **For non-exponential-family data, the relationship degrades by the Amari-Chentsov tensor (Open Problem 6.2).**

Explicit formulation:

Theorem (Information-Geometric Conditional Equivalence of Cercis, Revised)

Assumptions: All 8 assumptions (H1–H8) hold simultaneously.

Conclusion:

$$\mathcal{C} = \min_{d_0 h \in \text{im}(d_0)} \text{GeoDist}(P_A, P_{d_0 h})^2 = 2 \min_{d_0 h \in \text{im}(d_0)} \text{KL}(P_A \| P_{d_0 h}) + O(\|\mathbf{a}\|^3).$$

Cercis measures the minimal Fisher information distance from the observed expert edge distributions to the “pure-gauge bulk states” — the hypothetical world where all experts can be perfectly aligned.

Limitations:

- (i) The equivalence is rigorous within exponential families (H1); outside them, the difference between GeoDist^2 and 2KL is controlled by the Amari-Chentsov third-order tensor $T_{ijk} = \mathbb{E}[\partial_i \log p \cdot \partial_j \log p \cdot \partial_k \log p]$, which is generically non-zero for SCX data (Open Problem ??).
- (ii) The equivalence holds under local approximation (H2); for large discrepancies, third-order terms dominate.
- (iii) “Natural emergence” is an incorrect characterization — this is a *constructed equivalence* under strong assumptions, not an intrinsic geometric inevitability of the data.

4 / Five Breakdown Points

SCX

Below, we summarize five principal ways the equivalence can break on real SCX data:

B1: / Non-Exponential-Family Failure (Most Severe).

SCXMoEsoftmax $T(x)$ 6.2??

SCX expert outputs (MoE routing weights, softmax scores, attention distributions) are extremely unlikely to form an exponential family. Exponential families require specific algebraic structure (linear sufficient statistics $T(x)$) that deep learning outputs do not generally satisfy. The document itself lists this as Open Problem 6.2 (??).

B2: “KL” “KL” / Confusing “min KL” with “average KL”.

Cercis $\min_h \text{KL}(P_A \| P_{d_0 h})$ KL

Cercis solves $\min_h \text{KL}(P_A \| P_{d_0 h})$ — a minimization problem, not an average of KL divergences over all expert pairs. These are mathematically distinct.

B3: / Hidden Euclidean Approximation Requirement.

CercisFisher $\mathcal{I}_{ij} = \delta_{ij}$ (H7) — SCX

The equivalence between Euclidean-norm Cercis and Fisher geodesic distance requires $\mathcal{I}_{ij} = \delta_{ij}$ (H7) — meaning the information manifold must be everywhere flat, which is extremely unlikely on real SCX parameter spaces.

B4: 4.2 / Intrinsic Correction in Theorem 4.2 Proof.

$\text{im}(d_0) \ker(L_1)$??

Choosing the wrong submanifold loses the harmonic component. The correct definition is: the bulk submanifold is parameterized by exact connections (pure gauge) $\text{im}(d_0)$, not harmonic bulk states $\ker(L_1)$. Theorem ?? identifies and corrects this issue in its proof.

B5: / Locality Constraint.

$$\|\theta_q - \theta_p\| O(\|\Delta\theta\|^3)$$

All results hold only when $\|\theta_q - \theta_p\|$ is sufficiently small. When inter-configuration expert differences are large (e.g., different domains, different training stages), the third-order error term $O(\|\Delta\theta\|^3)$ is non-negligible.

5 / Recommended Revisions

5.1 / Changes Needed in `gauge_formalized.tex`

(1) **234–238** “”

“FisherKL...”

“FisherKL...”

(2) ??

“FisherKL / Fisher Geodesic–KL Equivalence under Exponential Families”

“FisherKL / Conditional Fisher Geodesic–KL Equivalence under Exponential Families”

(3) ?? H1–H8

(4) ??

“CercisFisher2KL”

“CercisFisher 2KL6.2”

(5) **1529–1532**

“Cercis: $\min_{d_{0h}} \text{GeoDist}(P_A, P_{d_{0h}})^2$ ” “Cercis: $\approx 2 \min_{d_{0h}} \text{KL}(P_A \| P_{d_{0h}})$ ”

“”

(6) ??“” **1560–1564**

“SCXS...”

“SCXS 6.2Amari-Chentsov”

(7) **31678–1684**

“FisherKL...Cercis”

“FisherKL...Cercis8”

6 6.2 / Cross-Reference to Open Problem 6.2

?? “Fisher-KL”

SCXFisherKL Amari-Chentsov $T_{ijk} = \mathbb{E}[\partial_i \log p \cdot \partial_j \log p \cdot \partial_k \log p]$ SCX

This corrigendum is directly linked to Open Problem ?? (“Fisher-KL Equivalence Outside Exponential Families”), which states:

For general SCX output distributions (potentially non-exponential-family), the relationship between Fisher geodesic distance and KL divergence is controlled by the Amari-Chentsov third-order tensor $T_{ijk} = \mathbb{E}[\partial_i \log p \cdot \partial_j \log p \cdot \partial_k \log p]$. Bounds on this tensor for empirical SCX output distributions are needed.

`docs/reviews/GAUGE_5VIEWPOINTS_FINAL.md`“” `docs/reviews/GAUGE_VIEWPOINTS_REVIEW.md` 4✓ “” “”
Review basis: This corrigendum is based on: `docs/reviews/GAUGE_5VIEWPOINTS_FINAL.md` (verdict: “overstated”) and `docs/reviews/GAUGE_VIEWPOINTS_REVIEW.md` (detailed analysis). The verdict on Viewpoint 4 is: ✓ mathematically correct relationship but “natural emergence” is severely overstated — downgrade to “conditional equivalence”.